The fdamador neutrino: a report

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Summary

Turning the toroidal photon's field inside out gives a neutral, self-sustaining ring of electric field, which matches two neutrino properties but cannot stay coherent long enough to be one. Hypothesis H28 is recorded as *bounded*.

- **What matches:** the inverted ring has zero charge and zero time-averaged magnetic moment, and its crossed circuit form keeps the half-integer (spin ½) modes.
- **What fails:** in vacuum it radiates away in about 3×10^{-16} s. SN1987A neutrinos survived about 5×10^{4} s of their own time, a gap of about 20 orders of magnitude.
- **The inflow:** the zero-point field of its own radiation resistance balances the outflow. The ring holds $\frac{1}{2}\hbar\omega_0$ forever with the energy ledger closed, but as random noise, not as a particle.
- **Missing entirely:** mass and the weak interaction, which is how every neutrino has been detected.
- **Implosion and explosion:** the ring takes in the exact time reverse of its ring-down whole (0.99966) and explodes it back out, but every implosion is paid for by an incoming wave from outside.
- **Flavours (H39):** travelling, standing and cancelling waves on the ring are stable and, with a defect, convert in the form of neutrino oscillation; tuned, they match the measured mixing angles, but the three types span only two states and carry no mass or weak interaction.

Every number below was computed in kronspice with an analytic twin beside it and a closed energy ledger.

The idea

The fdamador neutrino is the toroidal photon electron (Williamson and van der Mark, 1997) with its field inverted: the electric field forms the ring and the magnetic field circles round it. It was proposed by Franklin Amador on 4 November 2026 and entered in the kronspice hypothesis register as H28.

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In the photon electron and the charged rings before it, a current runs round the ring and B wraps the tube. Here E runs round the ring and B wraps it. The proposal is that this inverted structure is the neutrino, so it is tested against the neutrino′s measured properties rather than the electron′s.

What Maxwell's equations say before any model

Three results follow from the field equations alone:

1. **No charge.** A closed loop of E has ∇·E = 0, so by Gauss′s law it encloses no charge. It is neutral, as a neutrino is, and so it can never be an electron.
2. **No static form.** E can circulate round the ring only while the magnetic flux through the hole changes (Faraday′s law). The ring sustains itself as an oscillation: E round the ring is a ring of displacement current, its B threads the hole, and the changing flux drives E.
3. **An oscillating magnetic moment.** The displacement-current ring is a magnetic dipole whose time average is zero. A time-varying magnetic dipole radiates, so the ring loses energy.

This field pattern already exists in engineering: it is the TE01δ mode of a dielectric ring resonator, used in microwave filters, where the dielectric holds the field in place.

Part 1: the E-ring resonator

A 10 mm ceramic ring with εr = 100 resonates at 2.0606 GHz with Q = 40.5, and kronspice agrees with the formula to four figures by two independent methods.

The ring is a lumped loop of 24 sections: series capacitors for the displacement path along the ring, the ring′s self-inductance, a radiation resistance for its magnetic-dipole radiation, and dielectric loss. Radiation alone gives a closed-form Q:

```
Q_{rad} = \frac{6\Lambda}{\pi (ka)^3}, \qquad ka = \frac{\sqrt{2}\,(a/r)}
{\sqrt{\varepsilon_r \Lambda}}, \qquad \Lambda = \ln\frac{8a}{r} - \frac{7}
{4}
```

So Q grows as εr^1.5: **the permittivity is the confinement.** Bench ring: a = 10 mm, r = 2.5 mm, tan δ = 1×10⁻⁴.

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εr	f_0 (GHz)	ka	Q, formula	Q, kronspice AC
30	3.762	0.788	6.68	6.68
100	2.061	0.432	40.52	40.52
300	1.190	0.249	207	207
1000	0.652	0.137	1,140	1,140

At εr = 100 a ring-down gives Q = 40.53. Its ledger closes: the 3.151×10^{-13} J kick equals 3.144×10^{-13} J radiated and heated plus 6.7×10^{-16} J still stored, with a time-stepping error of 2.5×10^{-18} J. No charge leaves the ring, and the moment averages to −0.007 of its peak. At εr = 30 the ring is no longer small against the wavelength (ka ≈ 0.8), so that row sits at the edge of the lumped model.

Part 2: the neutrino test bench

To be a neutrino the ring would need an effective vacuum permittivity of 3.4×10^{13}, and nothing known supplies one.

In vacuum (εr = 1, a/r = 4) the ring is electrically large at its own resonance (ka = 4.3), so the lumped model fails and gives Q ≈ 0.04: the field is gone within one period. The field solve of the uninverted photon loop shows the same, with light leaving in under half a period.

The bench asks what holding it would take. SN1987A neutrinos (about 10 MeV) crossed 168,000 light-years. At a declared, illustrative mass of 0.1 eV the Lorentz factor is 10^8, so the proper time is 5.3×10^4 s and the ring must keep its energy for $Q = 8\times10^{18}$.

Property	E-ring model	Measured or standard
Charge	0 ($\nabla\cdot E = 0$)	consistent with 0
Spin ½	crossed inverted ring keeps half-integer modes (Part 3)	½
Static magnetic moment	0 (it oscillates)	< 2.8×10^{-11} μB (Borexino 2017); standard ≈ 3×10^{-20} μB at 0.1 eV
Lifetime in vacuum	≈ 3×10^{-16} s	⩾ 5.3×10^4 s proper time (SN1987A)

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Property	E-ring model	Measured or standard
Confinement to survive	effective $\varepsilon r = 3.4\times10^{13}$ at $a = 1.5\times10^{-12}$ m	none known
Mass	not given (0.1 eV declared)	< 0.45 eV (KATRIN 2025)
Weak interaction	not in the model	how every neutrino is detected

Part 3: the inverted travelling wave

Crossing the conductors of the inverted ring line still gives half-integer modes, so the spin-½ property of the photon electron survives the inversion.

Swapping E and B in a ring transmission line gives its circuit dual: series capacitors carry E along the path and shunt inductors carry B round it. That is a backward-wave ladder, where the longest wavelength has the highest frequency. Its modes follow $\omega = 1/(2\sqrt{(C\cdot Lg)}\cdot\sin(\pi k/N))$, with k whole on the plain ring and half-integer on the crossed one. Built in kronspice with 100 sections, C = 1 nF and Lg = 1 µH:

Mode	Plain ring (MHz)	Wavelengths	Crossed ring (MHz)	Wavelengths
1	80.11	1	160.21	½
2	40.08	2	53.42	1½
3	26.74	3	32.07	2½
4	20.08	4	22.93	3½

The AC peaks match the ladder formula to within 5×10^{-5}, and both AC ledgers balance.

Part 4: an inflow that balances the outflow

The zero-point field balances the ring's radiation exactly, so the ring never decays to zero, but what it holds is half a quantum of random noise.

By the fluctuation–dissipation theorem (Callen and Welton, 1951), the resistance that radiates the ring's energy is also a noise source: the field it radiates into drives it back. At the neutrino ring's 0.1 eV, thermal noise is negligible even in the 2.7 K cosmic background

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(occupation e^{-430}), leaving the zero-point part. Inflow and outflow are both booked on the magnetic-field channel, so they share one ledger column. Results from a scaled loop at the bench ring's radiation Q of 40.7, over 4 realizations:

Quantity	kronspice	Expected
Stored energy	$0.4941 \pm 0.0067\ \hbar\omega_0$	$0.4916\ \hbar\omega_0$ (½$\hbar\omega_0$ for the modes present)
Magnetic channel	391.77 J in = 389.36 J out + 2.15 J stored	in = out + stored
Energy fluctuation $\sigma/\langle E\rangle$	0.97	≈ 1 for a random field
Mean magnetic moment	1.9×10^{-4} of its rms	0

Does the Casimir effect add inflow? No, but the boundaries that cause it change the outflow. The Casimir force comes from walls changing which zero-point modes fit between them, a fixed energy shift rather than a continuous supply. The same walls change the ring's radiation rate by a Purcell factor F, and the noise changes with it, so the held energy stays put:

Purcell factor F	Q	Stored, kronspice ($\hbar\omega_0$)	Expected ($\hbar\omega_0$)	Lifetime
4 (enhanced)	10.2	0.4794 ± 0.0023	0.4817	×¼
1 (free space)	40.7	0.4941 ± 0.0067	0.4916	×1
0.2 (suppressed)	203	0.4843 ± 0.0318	0.4942	×5

A coherent ring lasting SN1987A's proper time would need $F \approx 5\times10^{-21}$. The classic measurement of inhibited emission found about 20× (Hulet, Hilfer and Kleppner, 1985), and a free neutrino has no walls.

What the expected value caught. The first version read 8 % high at low Q. The noise started at a non-zero voltage, so the start-up operating point charged the series capacitors through 1 GΩ bleeders, an offset that relaxes over 4×10^{9} s and biased the stored energy in proportion to R_rad. Ramping the noise on from zero removed it, and

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both points then agreed within 0.5σ. The ledger itself stayed closed, because the offset was booked as energy stored at the start; the expected value computed beside the result is what exposed it.

Because the ramp was added after the mismatch was seen, it was tested for whether it steers the result: on identical noise, ramps of 2Q, 5Q and 8Q agree to four digits, and an abrupt start reproduces the bias exactly. The full account, with the equations, the record of events and the check, is in Reference: the 8 % zero-point bias.

Part 5: implosion and explosion

The ring has both halves, but the implosion must be supplied from outside: only the exact time reverse of its ring-down is taken in whole (0.99966), and then it explodes as the ordinary ring-down. This tests H30 (a particle as a cycle of contraction and expansion) on the E-ring. In standard physics the inward half of a radiating resonator is its ring-down run backwards, an incoming wave growing as e^{+αt} at f_0; every joule of it is declared and booked.

Circuit (kronspice). The Part 1 bench ring (2.0606 GHz, Q 40.52), with its radiation resistance R_rad treated as a port to the outside. Every drive carries the same incident energy, 2.298×10^{-10} J, over 8 τ_E (τ_E = Q/ω_0 = 3.129 ns). "Absorbed" is the share not sent back while the drive lasts.

Incoming wave	Absorbed	Peak stored / incident	Explosion / incident
Time-reversed ring-down (coherent perfect absorption rate)	0.999656	0.99536	0.99123
Same shape, twice the rate	0.889452	0.88742	0.88373
Same shape, half the rate	0.901874	0.89483	0.89112
Constant burst at f_0, 2 τ_E	0.799747	0.79681	0.79350
Constant burst at f_0, 8 τ_E	0.486970	0.47658	0.47459
The ring-down itself, not reversed	0.028366	0.53939	0.02031

- The time-reversed drive's whole reflection, 3.4×10^{-4}, is its switch-on (it starts at e^{-8} of its final power). This is coherent perfect absorption, the time-reversed laser (Chong, Ge, Cao and Stone, *PRL* 105, 053901, 2010).

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- Ledgers close to 6.7×10^{-6}. CPA run: incident 2.29791×10^{-10} J = reflected 7.914×10^{-14} J + explosion 2.27775×10^{-10} J + heat 1.857×10^{-12} J + stored 7.77×10^{-14} J.
- **Pulsing ring.** With one implosion every 40 periods, the steady cycle takes in 2.175×10^{-10} J and gives back 2.158×10^{-10} J. The difference, 1.718×10^{-12} J (0.79 %), is the dielectric heat (1.716×10^{-12} J). A drive in antiphase settles to the same cycle by cycle 4.

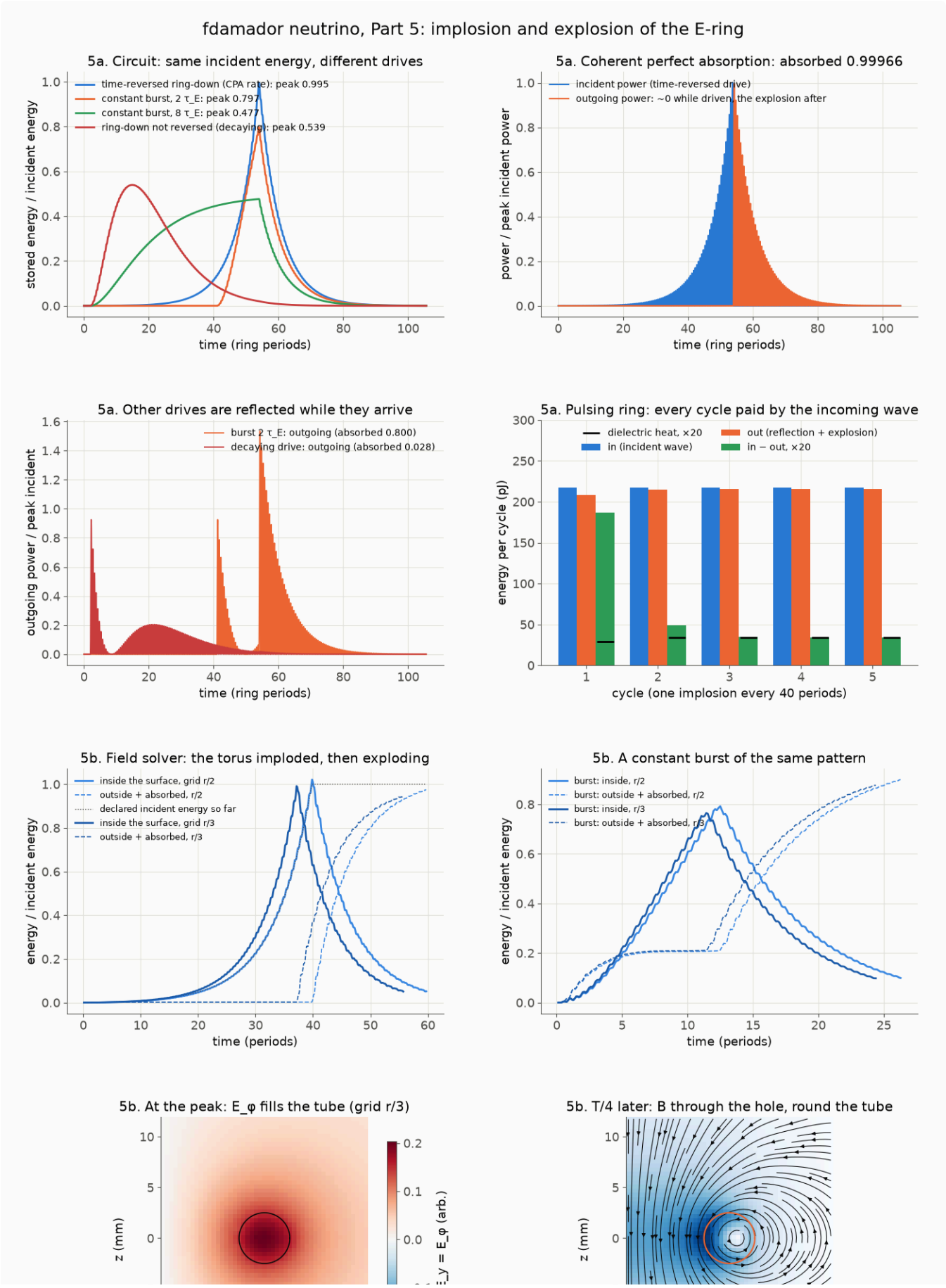
Field (em/fdtd.go). The same torus (εr = 100, lossless) in the field solver, inside a closed Huygens surface at ±2a. The surface feeds in the time reverse of the ring's own outgoing field, growing at the ring-down rate, then switches off at an E maximum.

Quantity	Grid r/2	Grid r/3	Circuit
f_0	2.101 GHz	2.0571 GHz	2.0606 GHz
Q	41.69	38.9	Q_rad 40.68
Held at the peak / delivered	0.9976	0.9974	0.99536 of the incident
Constant burst, peak held	0.7923	0.7649	0.79681
E round the ring: share of the electric energy at the peak	0.9966	0.9971	1
Correlation with the free ring-down's E	0.9991	0.9996	—
Field ledger residual / incident	3.3×10^{-13}	4.8×10^{-13}	5.1×10^{-6}

The lumped f_0 and Q lie between the two grids (f_0 moves 2 % and Q 7 % between them; a finer r/4 grid was not run). At the peak the rebuilt field is the TE01δ pattern: E round the ring, and a quarter period later B through the hole, closing round the tube. After the peak 0.947 of the delivered energy explodes outward within 3 τ_E.

Standard prediction, borne out by both: the implosion is the explosion reversed, absorbed without reflection only when it is the exact time reverse, and every cycle is paid for by the incoming wave. Nothing in the ring supplies that wave; for a neutrino it would have to arrive from outside, coherent, every period.

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Part 5 in kronspice and the field solver: the same incident energy under different drives; coherent perfect absorption (incident in, nothing out until the explosion); the pulsing ring's books per cycle; the torus imploded and exploding on grids r/2 and r/3; at the peak E fills the tube round the ring, and a quarter period later B threads the hole.

Part 6: flavours as wave types (H39)

The three wave types are stable states of the ring and, with a defect added, they convert into one another in the same form as neutrino oscillation; tuned, the ring reproduces the measured mixing angles, but that is a fit, and it has nothing of what defines a flavour. The hypothesis, entered as H39 on 2026-11-07: the three neutrino flavours are three kinds of wave on the ring, constructive, destructive and standing.

Where the types live. The uniform E-ring (TE01δ) is a single state with no travelling or standing form. The types live on the ring's harmonics that vary round it, built as the inverted ring lines of Part 3 (24 sections). Each harmonic is a pair of waves running opposite ways, A (clockwise) and B (anticlockwise). One fixed definition, taken at a reference point c:

Type	Definition	What it means
C, constructive	travelling, B = 0	every point reaches the full amplitude
S, standing	|A| = |B|, antinode at c	fixed nodes and antinodes
D, destructive	|A| = |B|, node at c	the two waves cancel at c

No state cancels everywhere: the energy is $N(|A|^2 + |B|^2)$ exactly, so only zero does. And $C = (S + iD)/\sqrt{2}$: **the three types span two dimensions, not three.**

Test 1, persistence. Each type, imposed on a perfect ring and followed for 100 periods in kronspice, keeps its own share to 1.0000000000 on both the plain and the crossed ring. C turns once a period (0.99909 turns), S and D keep their nodes, and the energy falls only by the shunt loss (ledgers about 10^{-4}). None decays or converts.

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Test 2, conversion. A capacitor defect δ on one section splits the pair, and a wave started clockwise turns anticlockwise and back: $P = \sin^2 2\theta \cdot \sin^2(\pi \Delta f \cdot t)$, the form of two-flavour oscillation.

Defect δ	Δf, exact	Δf, kronspice	$\sin^2 2\theta$
0.05	37,428 Hz	37,327 Hz	1.0000
0.10	71,454 Hz	71,266 Hz	1.0000
0.20	131,000 Hz	130,660 Hz	1.0001
0.40	224,530 Hz	223,990 Hz	1.0001

Started as a standing wave s sections from the defect, the angle is $\theta = \kappa \cdot s$: kronspice gives $\sin^2 2\theta = 0.2507$, 0.7512 and 1.0000 against 0.25, 0.75 and 1.

Three flavours need a third state. The ring's band-edge π-mode lies 0.86 % below the pair, and two defects mix the three. Tuned for the closest match to the measured mixing (NuFIT 6.0, normal ordering with Super-Kamiokande atmospheric data):

Quantity	Ring, tuned	Measured
Defects	δ = 0.2064 on section 2, 0.1628 on section 23	—
θ_{12}	33.66°	33.68° ± 0.72
θ_{23}	42.28°	43.3° ± 0.9
θ_{13}	8.56°	8.56° ± 0.11
Splitting ratio	33.56	$\Delta m^2_{31}/\Delta m^2_{21} = 33.55$
CP phase	0 or 180° only	212° (+26/−41)

That is four numbers fitted with four knobs, not a prediction. Up to 8.7 % of each mode lies outside the three ring states; kronspice follows the ring's full 24-mode expansion to 1.1×10^{-4} but departs from the 3×3 formula by up to 0.277.

Test 3, implosion and explosion of the coherent unit. A port on the ring at the reference point, with the same external Q as the E-ring (40.5):

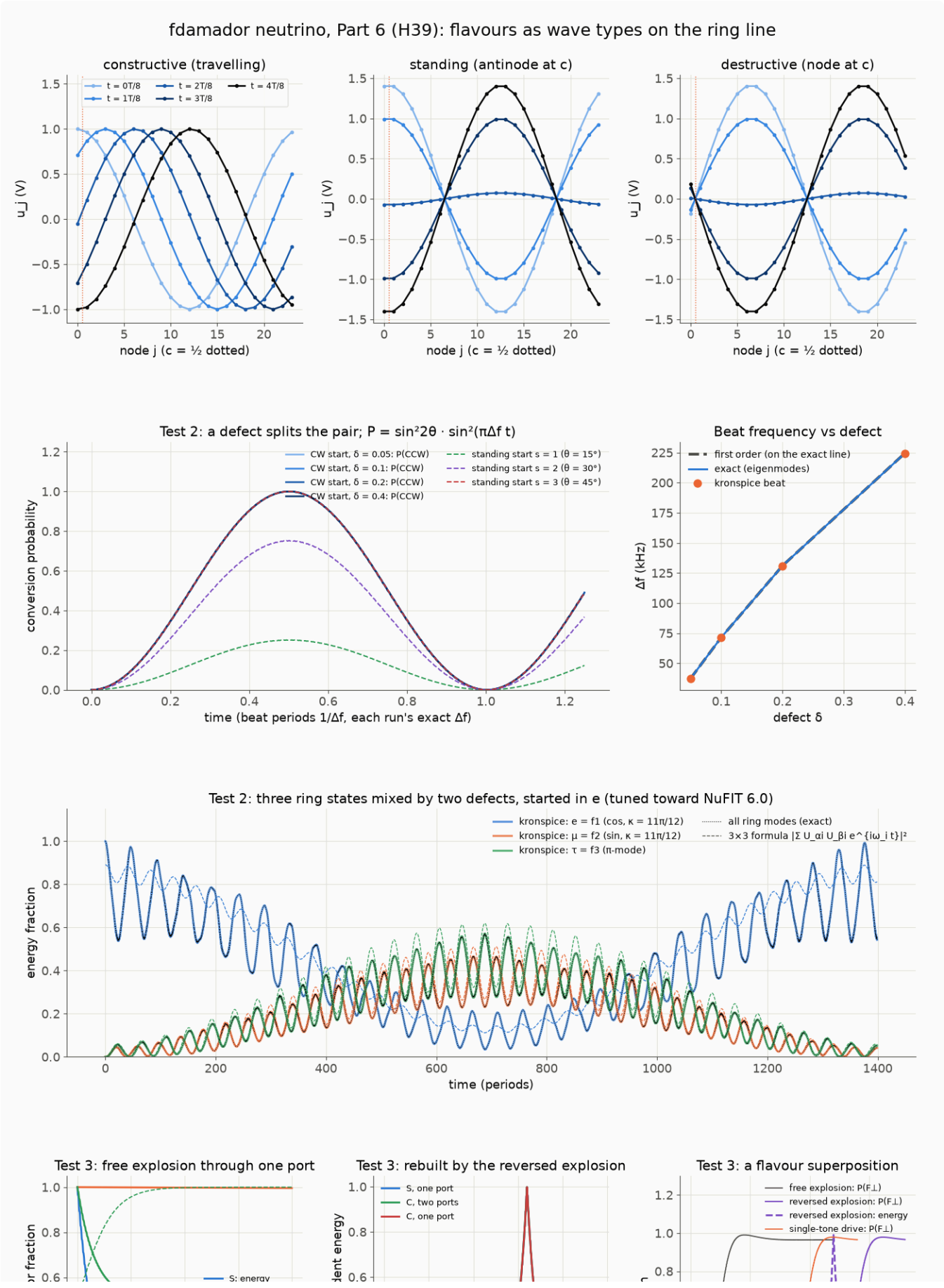
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Type	Free explosion: energy left	State left	Rebuilt by the reversed explosion: fidelity
S (antinode at the port)	0.0000	—	1.00000 (one port)
D (node at the port)	0.9957	D	cannot be built through this port
C (travelling)	0.4979	D	1.00000 (two ports in quadrature); 0.50000 (one port)
Flavour mixture, θ = 22.5°	0.0221	the beat continues	0.98440 (a single tone: 0.96903)

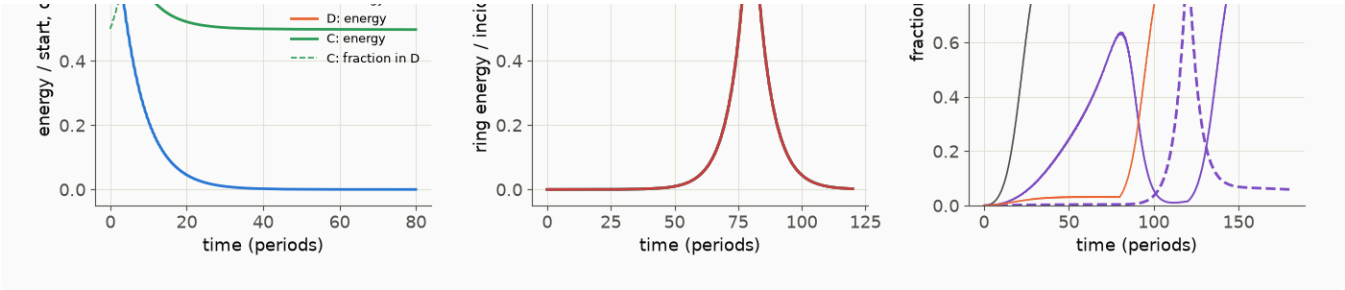
- The explosion changes the type: a standing wave with its antinode at the port leaves completely, one with its node there cannot leave, and a travelling wave loses half and is left standing.
- The implosion rebuilds any type it can reach, but only when the incoming wave is the exact reverse; a travelling wave needs two ports, and a flavour mixture needs both mode frequencies in the right phases.
- A flavour mixture keeps beating through the explosion, and the beat resumes after the rebuild.

Standard prediction: every combination of a degenerate pair is stationary, and the oscillation formula is the generic mathematics of coupled modes. What does not match: a neutrino's flavour is set by the weak interaction (the charged lepton it makes or is made with), its splittings are mass differences entering as $\Delta m^2 L/4E$, matter changes them (MSW), and a CP phase may be present. The ring has no weak production or detection, no mass, no matter effect, and a reciprocal lossless ring cannot carry a CP phase. H39 is recorded as *bounded*.

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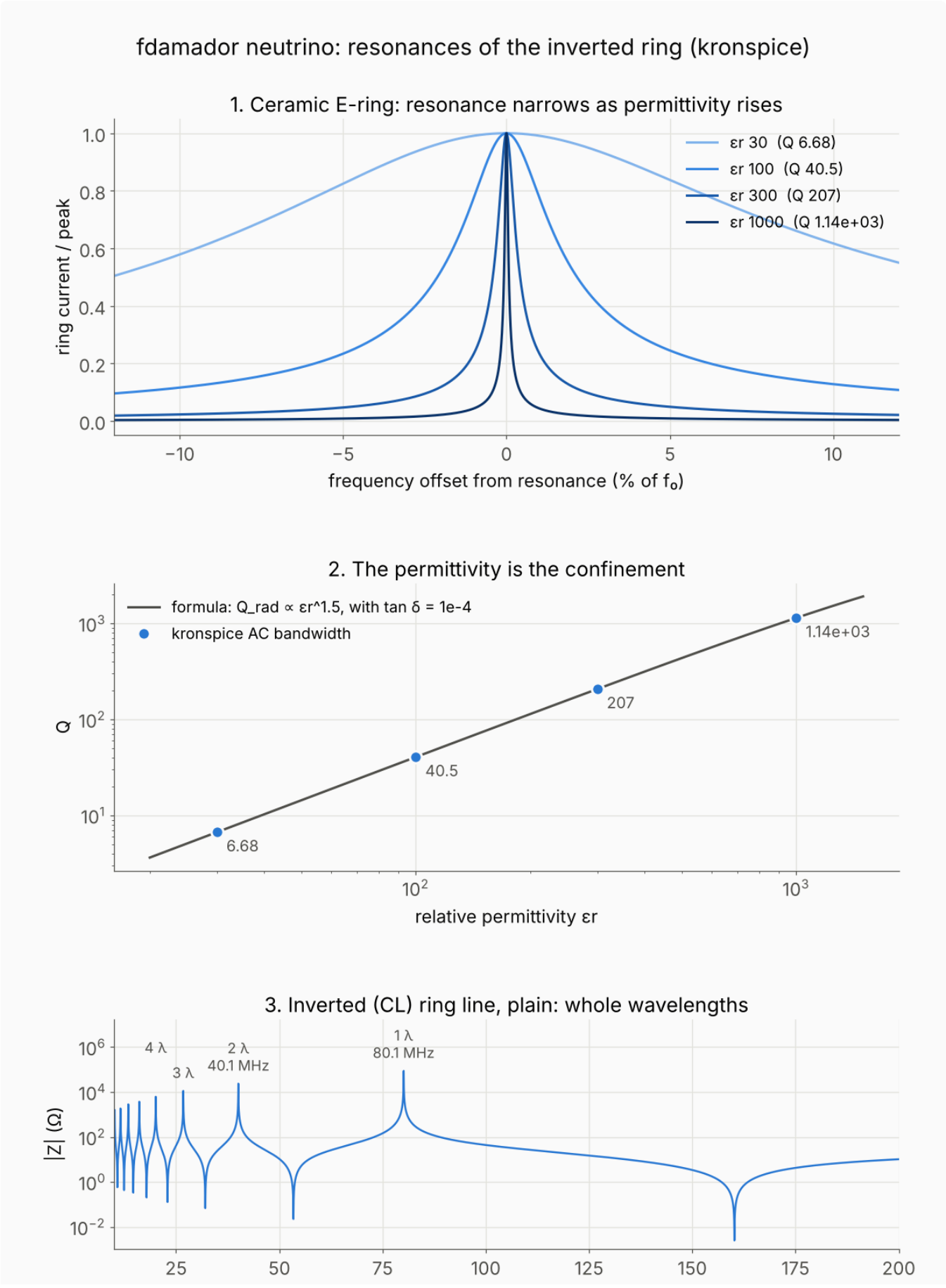


Part 6 in kronspice: the three wave types on the ring; the beat a defect makes, against δ; three ring states mixed by two defects tuned toward NuFIT 6.0 (kronspice against the full mode expansion and the 3×3 formula); each type's free explosion, its rebuild by the reversed explosion, and a flavour mixture through the cycle.

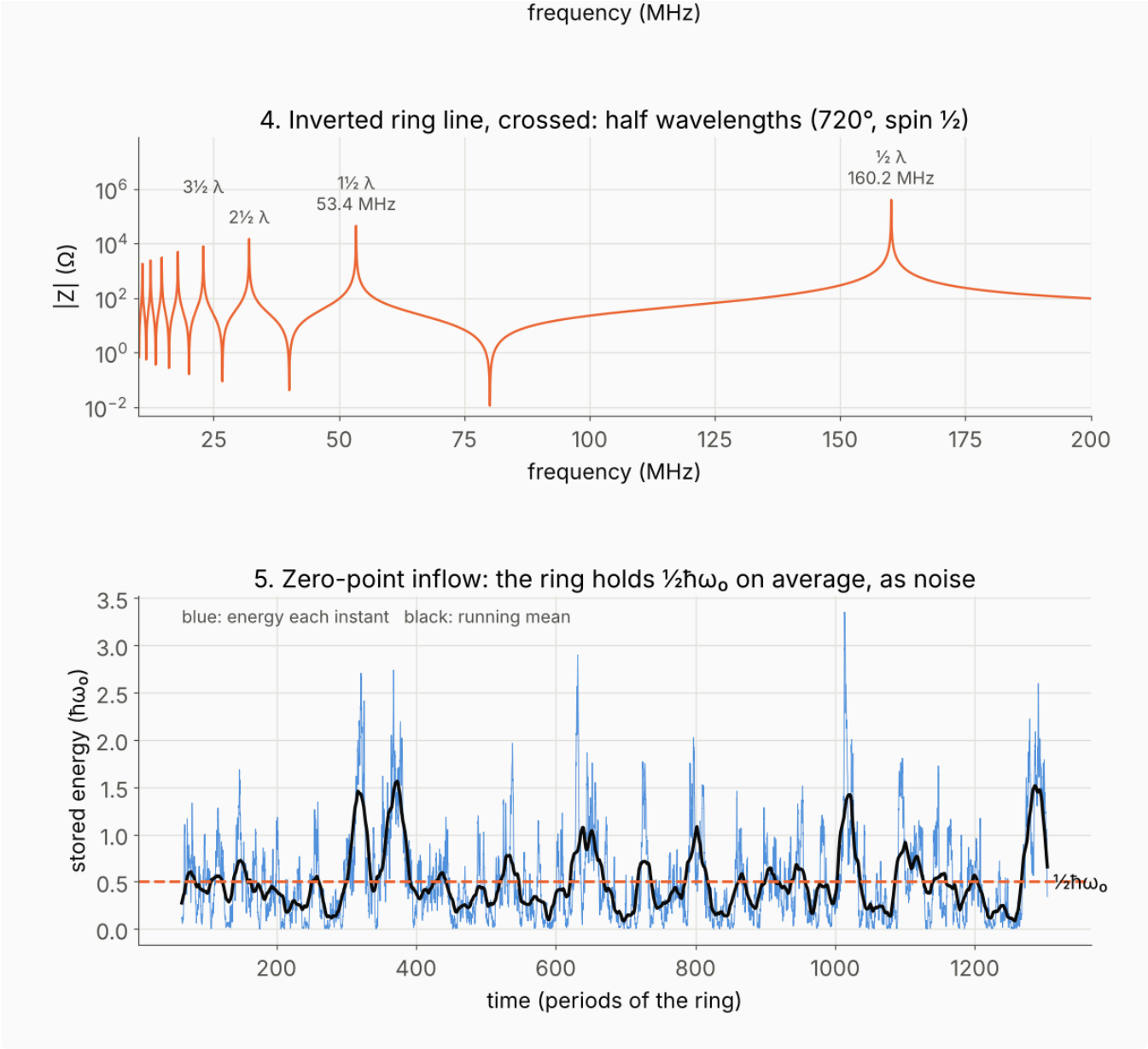
Chart and sound

The resonance chart shows five things: the E-ring's resonance narrowing as permittivity rises, Q on its εr^1.5 line, the plain and crossed ring-line spectra with each mode labelled in wavelengths, and one zero-point run with the stored energy fluctuating about ½ħω$_0$.

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Every real frequency here is far above hearing (2.06 GHz, 20–160 MHz, 24 THz), so each sound is the kronspice waveform slowed by one factor. Every frequency ratio, decay rate and Q is kept.

Sound	Source	What you hear
1. Ceramic ring, εr 100 (Q 40.5)	kronspice ring-down	a 0.14 s tick at 440 Hz
2. Ceramic ring, εr 1000 (Q 1,140)	kronspice ring-down	a 3.4 s bell at 440 Hz

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Sound	Source	What you hear
3. Ring held by the zero-point field	kronspice noise run	a 2.8 s hiss centred on 440 Hz
4. Plain inverted ring line	four highest modes, synthesized	880, 440, 294, 221 Hz: the undertones 1, ½, ⅓, ¼
5. Crossed inverted ring line	four highest modes, synthesized	880, 293, 176, 126 Hz: the odd undertones 1, ⅓, ⅕, ⅐

The harmonics file adds the ring-line modes one at a time, so each tone can be heard entering. The crossing removes every even member of the series: the spin-½ property heard rather than seen. Sounds 1–3 have no overtones, because each is a single linear resonance.

Conclusions and next steps

The inversion gets the two properties a field picture can get, no charge and no static moment, and keeps spin ½ in its circuit form; it fails on stability and has nothing to say about mass or the weak interaction.

The zero-point inflow closes the energy books but does not rescue the model. It holds every mode of every oscillator at $\frac{1}{2}\hbar\omega_0$, so it singles out no particle, and its fluctuations are as large as its mean.

Parts 5 and 6 add two results. The ring has an implosion as well as an explosion, but only an incoming wave from outside, the exact reverse of its ring-down, supplies it. And the wave types on the ring reproduce the form of flavour oscillation, and when tuned its angles, without the mass, weak interaction or CP phase that make a neutrino's flavour; H39 is recorded as bounded.

What would move H28 off *bounded*:

- [ ] A confining mechanism that suppresses the ring's radiation by about 5×10^{-21}, with its own equations and a test
- [ ] An account of where the neutrino's mass comes from in the ring picture
- [ ] A coupling to the weak interaction, the only way neutrinos have ever been detected
- [ ] A prediction that differs from the standard neutrino, measurable with existing data

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Repository and sources

The model lives in `examples/fdamador_neutrino` of fdamador/kronspice, on the local branch `fdamador-neutrino` (commits 4bfa244, a3ce7ba and 9e2deba; not yet pushed). All tests pass, including the zero-point inflow test. Run it with `go run ./examples/fdamador_neutrino`, and draw the chart and sound with `-viz` then `viz/render.py`.

Sources:

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